

Zainab Verstraeten

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EDUCATION

- University of Cambridge** | *MSc Computer Science* 2019-2020
- Achieved Distinction in thesis project on distributed systems optimization, improving cache efficiency by 34%
 - Led a team of 4 students in designing a containerized microservices architecture workshop attended by 120+ participants
- University of Amsterdam** | *BSc Computer Science* 2015-2019
- Graduated with Honors (GPA 3.8/4.0), specializing in systems programming and algorithms
 - Ranked top 5% in Data Structures course, mentored 8 junior students in advanced algorithm design

EXPERIENCE

- Cloudflare** | *Software Engineer II* 2022-Present
- Architected and deployed a real-time DNS query analytics pipeline processing 2.3 million requests/second, reducing query latency by 28% for enterprise customers
 - Led migration of legacy rate-limiting service from Python to Rust, decreasing memory footprint by 61% and improving throughput by 4.2x
 - Mentored 3 junior engineers on edge computing best practices, resulting in 2 successful production deployments and 100% code review approval rate
- Notion** | *Mid-Level Software Engineer* 2021-2022
- Optimized database query performance for collaborative editing features, reducing P99 latency from 850ms to 120ms and improving user retention by 15%
 - Implemented real-time synchronization protocol for offline-first functionality, enabling 40,000+ concurrent users with <200ms sync latency
 - Refactored permission system using Redis caching, cutting database load by 52% and saving \$47,000 annually in infrastructure costs
- Shopify** | *Junior Software Engineer* 2020-2021
- Developed payment reconciliation system processing \$89M+ in transactions monthly with 99.98% accuracy, eliminating 200+ manual reconciliation hours/month
 - Enhanced checkout page performance by optimizing React components and implementing code splitting, increasing conversion rate by 8.3%
 - Built comprehensive monitoring dashboard using Datadog and custom metrics, reducing incident detection time by 65%

PROJECTS

- StreamSync** | *Rust, WebSocket, Redis, PostgreSQL* 2023-Present
- Engineered a high-throughput streaming data synchronization library supporting 50,000+ messages/second with sub-100ms latency across distributed systems
 - Achieved 94% test coverage with comprehensive benchmarking suite, resulting in 3 open-source contributions and 1,200+ GitHub stars
- GraphQL Query Optimizer** | *TypeScript, Node.js, GraphQL, Jest* 2022-2023
- Developed an intelligent query optimizer reducing API response times by 46% through automated batch request detection and N+1 query prevention
 - Created educational documentation attracting 8,000+ developers, with 350+ upvotes on Hacker News and featured in 5 tech newsletters
- DecentralAuth** | *Go, Ethereum, Solidity, Docker* 2021-2022
- Built decentralized authentication middleware using smart contracts, processing 15,000+ authentications weekly with zero security incidents
 - Published peer-reviewed whitepaper and secured speaking slot at 2 blockchain conferences, reaching 5,000+ audience members
- CloudMonitor CLI** | *Python, AWS SDK, Click, PostgreSQL* 2020-2021
- Created open-source CLI tool for AWS resource monitoring, automating infrastructure audits for 2,000+ users and saving 1,500+ engineering hours quarterly
 - Achieved 12,000+ downloads on PyPI with 4.8/5.0 star rating and active community of 45+ contributors

EXTRACURRICULAR

- European Developer Conference (EuroDev)** | *Conference Speaker & Workshop Facilitator* 2022-Present
- Delivered 2 keynote talks on 'Scaling Distributed Systems at Edge' reaching 3,200+ engineers from 45 countries

- Facilitated hands-on workshop on performance optimization techniques, achieving 9.2/10 participant satisfaction score with 95% attendance rate

Open Source Collective (Apache Foundation Member) | *Core Contributor & Maintainer* 2021-Present

- Maintain 3 open-source projects with 18,000+ combined GitHub stars, reviewing 200+ pull requests annually and growing contributor base by 140%
- Mentor 5 junior developers transitioning to open-source contributions, with 100% of mentees successfully maintaining their own projects

TECHNICAL SKILLS

Languages – Rust, Go, TypeScript, Python, Java, SQL, Solidity

Technologies – PostgreSQL, Redis, Elasticsearch, GraphQL, React, Node.js, WebSocket, Apache Kafka, Docker, Kubernetes, AWS, Datadog

Tools – Git, GitHub, GitLab, VS Code, IntelliJ IDEA, Jira, Figma, Jenkins, Terraform, Linux, macOS